

Gate-Aware Renewable-Aligned Stochastic Battery Co-Optimization in DK1: Forecasts, Decisions, and Settlement Evidence

Abstract

This study evaluates a 4 MW/8 MWh price-taking battery in the sequential DK1 day-ahead energy and upward-aFRR markets. We propose Gate-aware Renewable-Alignment Stochastic Co-Optimization (GRASCO), which links gate-conditioned market scenarios, an ECMWF-driven weather-load proxy forecast, two-gate battery scheduling, and ex-post 15-minute settlement. On 211 delivery days, the proposed joint market scenarios significantly improve the Energy Score over independent quantile scenarios ($p = 0.0016$), but do not significantly outperform empirical or published block-copula comparators. Historical paired scenarios increase realized profit over LightGBM point dispatch by EUR 330.6/day and reduce the daily undelivered-activation share by 1.462 percentage points. In 330 complete proxy days, the VRE-above-load proxy is positive in 2,097 of 7,920 hours; every positive hour is also a net-export hour, so the proxy is not measured curtailment. WLPF identifies proxy-positive hours with 79.19% precision and 61.92% recall at 07:30, rising to 80.66% and 66.10% at 12:00. On 206 matched days, GRASCO-25, GRASCO-50, and GRASCO-100 monotonically increase midday charging, realized proxy overlap, and 17:00 SOC. Relative to GRASCO-0, realized profit changes by +13.4, −18.0, and −71.5 EUR/day, while evening reserve and delivery remain statistically unchanged because reserve is fixed at 07:30 and already close to the battery power limit. The results support probabilistic scheduling and renewable-rich charge alignment, but not a claim of actual curtailment reduction.

1 Research question and contribution

The research question is whether a price-taking battery can use information available at the two DK1 market gates to align midday charging with renewable-rich conditions while preserving profitable and deliverable evening upward aFRR. The contribution is a complete data-to-settlement experiment, not a claim that joint energy-reserve optimization, stochastic programming, or CVaR is itself new. Those ingredients are established in the storage and sequential-market literature [13, 14, 15, 10].

The method proposed in this paper is named *Gate-aware Renewable-Alignment Stochastic Co-Optimization* (GRASCO). GRASCO combines two models developed for this study: *Gate-Conditioned Joint Scenarios* (GCJS) for market uncertainty and a *Weather-Load Proxy Forecast* (WLPF) for the renewable-rich signal. The implemented renewable-alignment experiment uses the empirically stronger pooled Gaussian-copula market scenarios as its frozen market input; hence GRASCO denotes the complete two-gate decision architecture and renewable-alignment mechanism, rather than one particular copula estimator. Its experimental variants are GRASCO-0, GRASCO-25, GRASCO-50, and GRASCO-100, where the suffix is the alignment credit ρ in EUR/MWh. GRASCO-0 is the market-only control; the positive values are pre-specified sensitivity arms. The full optimization model is stated in Section 10, and WLPF is stated in Section 9.

The paper makes three distinctions throughout. First, point forecasts, probabilistic scenarios, and battery decisions are different objects. Second, methods reproduced from their original

descriptions are labelled as *DK1 reconstructions* and are cited to their source papers; they are not reported as replications of the authors’ original numerical studies. Third, the observed VRE-above-load quantity is a reproducible diagnostic proxy, not measured curtailment. These distinctions determine the method taxonomy in Table 5 and the admissible claims in Table 20.

How to read this paper. Sections 9–10 and the settlement trace that follows them define the complete GRASCO model; a reader who only wants the findings can read the notation in Table 1 and Table 3 and then skip directly to Section 13 below. The four result blocks that carry the paper’s main claim are: decision value over point dispatch (Table 12), the scale and accounting composition of the DK1 VRE-above-load proxy (Tables 14–15), the midday-charging and evening-reserve linkage by method (Table 17), and the GRASCO alignment-credit sensitivity (Table 18). Sections 9 and 10 and the intervening forecast-construction and settlement-trace sections supply the equations that produce those four result blocks and can be consulted only when a specific quantity needs to be traced to its formula; Table 7 indexes that mapping directly.

2 Scope and terminology

This section distinguishes system-level renewable energy utilization from ex-post proxy metrics that can be evaluated using the available DK1 data and simulated battery schedules. The repository-wide terminology and evidence rules are defined in `CONVENTIONS.md`. Variable renewable energy (VRE) refers only to wind and solar generation. The DK1 data do not contain unconstrained available VRE generation, curtailment instructions, observed curtailed energy, or source tracing for battery charging. Consequently, the DK1 proxy metrics defined below must not be reported as actual curtailment reduction or as an observed renewable utilization rate.

Let $t \in \mathcal{T}$ denote a 15-minute settlement interval and let $\Delta t = 0.25$ h. UTC timestamps remain the unique data keys and are mapped to Europe/Copenhagen local delivery time only when defining reporting windows. For delivery day d , the midday and evening-peak sets are fixed as

$$\mathcal{M}_d = \{t : t \text{ falls between 10:00 and 15:59 on day } d\}, \quad (1)$$

$$\mathcal{E}_d = \{t : t \text{ falls between 17:00 and 21:59 on day } d\}. \quad (2)$$

All UTC observations are retained on daylight-saving transition days; repeated local clock times are not deduplicated.

Two audited facts motivate fixing $\mathcal{M}_d$ and $\mathcal{E}_d$ at these clock hours. Across 335 audited days, the day-ahead price averages 64.52 EUR/MWh in $\mathcal{M}_d$ against 129.10 EUR/MWh in $\mathcal{E}_d$, and the upward-aFRR capacity price averages 9.84 EUR/MW against 36.86 EUR/MW; the evening day-ahead price exceeds the midday price on 89.9% of days, and the evening capacity price exceeds the midday price on 93.7% of days. Over the same days, the mean wind-plus-solar-to-load ratio falls from 0.800 in $\mathcal{M}_d$ to 0.713 in $\mathcal{E}_d$, and across 331 complete generation-and-load days the day-level correlation between this midday-to-evening decline in the VRE-to-load ratio and the corresponding rise in net load is 0.923 (authors’ calculations from Energinet settlement data [16]). These are audited descriptive statistics, not a pre-registered confirmatory test; they motivate, but do not by themselves establish, the decision results reported in Section 13.

3 Indices, information sets, and dependency graph

The mathematical objects use the indices in Table 1. The map $h(t)$ is required whenever an hourly quantity is used in a 15-minute equation. For an hourly power series Y_h , the notation $Y_t := Y_{h(t)}$ denotes a broadcast to the four constituent settlement intervals; energy is formed only after multiplication by Δt . No interpolation is performed by this mapping.

Table 1: Indices and deterministic maps used by the formula chain.

Symbol	Domain	Role
d	$\mathcal{D}$	delivery day
t	$\mathcal{T}_d$	15-minute settlement interval
h	$\mathcal{H}_d$	delivery hour
$h(t)$	$\mathcal{T}_d \rightarrow \mathcal{H}_d$	interval-to-hour map
s	$\mathcal{S}_d$	equal-weight market scenario
c	$\mathcal{C}_d$	07:30 capacity-price-path cluster
$c(s)$	$\mathcal{S}_d \rightarrow \mathcal{C}_d$	scenario-to-cluster map
v, b, k	$\mathcal{V}, \mathcal{B}, \mathcal{K}$	VRE technology, border, non-VRE category

Observed procured upward-aFRR capacity is $Q_{d,h}^{\text{proc}}$ in MW, and observed activated upward energy is $A_{d,t}^{\text{up}}$ in MWh per 15-minute interval. The realized activation-energy coefficient is

$$a_{d,t}^{\text{obs}} = \frac{A_{d,t}^{\text{up}}}{Q_{d,h(t)}^{\text{proc}}}, \quad Q_{d,h(t)}^{\text{proc}} > 0, \quad (3)$$

with units MWh per MW of reserve. Its scenario counterpart $a_{d,s,t}$ converts an hourly reserve commitment into requested interval energy. The scenario probabilities satisfy $\pi_s \geq 0$ and $\sum_s \pi_s = 1$; the saved experiment uses $|\mathcal{S}_d| = 50$ and $\pi_s = 1/50$.

Table 2 separates quantities that are otherwise easy to conflate. A hat denotes a gate-time forecast, “obs” denotes a realized observation, “plan” denotes a pre-activation state, and “real” denotes a scenario or ex-post state after activation and paid recovery.

Table 2: Inputs, outputs, and downstream use in the implemented analysis.

Stage	Output	Downstream use
Realized DK1 wind, solar, and load	X^{proxy}	ex-post decomposition and alignment diagnostics only
Gate-visible weather and published lags	$\hat{X}^{\text{proxy},g}$	forecast evaluation and the proxy-aware experimental decision arms
Gate-visible market history	$\hat{\Xi}_{d,s}^{(g)}$	reserve, energy, SOC, delivery, and recovery optimization
07:30 stochastic programme	common $r_{d,h}^{\text{up}}$	frozen input to the 12:00 programme
12:00 programme	$P^{\text{ch}}, P^{\text{dis}}, S^{\text{plan}}$	deployed energy schedule and planned SOC
Realized activation and prices	$S^{\text{real}}, z_{d,t}$	settlement and daily diagnostic vector J_d

4 Data-to-settlement research chain

The implemented analysis has three connected branches rather than one linear sequence:

$$\mathcal{D}_{<d,g}^{\text{market}} \rightarrow \hat{\Xi}_{d,s}^{(g)} \rightarrow u_d \rightarrow z_{d,t} \rightarrow J_d, \quad (\text{C1})$$

$$\mathcal{D}_{<d,g}^{\text{weather+lag}} \rightarrow \hat{X}_{d,h}^{\text{proxy},g} \rightarrow \left\{ \text{forecast scores}, u_d^{(\rho)} \right\}, \quad (\text{C2})$$

$$\mathcal{D}_{d,h}^{\text{proxy,obs}} \rightarrow X_{d,h}^{\text{proxy}} \rightarrow J_d. \quad (\text{C3})$$

Here, $\mathcal{D}_{<d,g}$ contains only information visible by gate g . Chain (C1) is the deployed market-decision chain. Chain (C2) measures whether a gate-safe renewable-rich signal exists and supplies

it to the proxy-aware experimental arms. The market-only baseline is recovered at $\rho = 0$, while $\rho > 0$ allows the gate-visible signal to affect charge timing. Chain (C3) uses realized wind, solar, and load only after delivery; the realized proxy is never supplied to either decision gate.

The forecasting target vector for scenario s is

$$\hat{\Xi}_{d,s}^{(g)} = \left\{ \hat{\lambda}_{d,s,t}^{\text{DA},(g)}, \hat{\kappa}_{d,s,h}^{\text{up},(g)}, \hat{a}_{d,s,t}^{(g)}, \hat{\lambda}_{d,s,t}^{\text{act},(g)}, \hat{\lambda}_{d,s,t}^{\text{imb},(g)} \right\}, \quad (4)$$

where λ^{DA} is the day-ahead energy price, κ^{up} is the upward-aFRR capacity price, a is the activation-energy coefficient defined by Eq. (3), λ^{act} is the activation price, and λ^{imb} is the imbalance price used for shortfall and recovery settlement. The experiment uses 50 equal-weight scenarios. The market vector in Eq. (4) is read from the frozen S3 scenario archives, whose construction is stated in Section 8; it is not generated by the weather-load regression in Chain (C2). The former drives the optimizer, while the latter forecasts only the diagnostic proxy. The 07:30 solve chooses the hourly reserve vector. At 12:00 the realized capacity price and updated energy-side information are introduced, while the 07:30 reserve commitment remains frozen. The 12:00 programme first maximizes expected activation delivery and then maximizes the risk-adjusted profit within a 10^{-9} MWh delivery tolerance.

The observed DK1 inputs and their native resolutions are

$$\mathcal{D}^{\text{market}} = \{\lambda_t^{\text{DA}}, \kappa_h^{\text{up}}, Q_h^{\text{proc}}, A_t^{\text{up}}, \lambda_t^{\text{act}}, \lambda_t^{\text{imb}}\}, \quad (5)$$

$$\mathcal{D}^{\text{proxy}} = \{P_h^{\text{wind,obs}}, P_h^{\text{solar,obs}}, L_h^{\text{obs}}, F_{b,h}, C_{k,h}, H_h\}. \quad (6)$$

Here, $F_{b,h}$ is the signed exchange on border b (positive for imports), $C_{k,h}$ is generation in non-VRE category k , and H_h is observed Power-to-Heat consumption. Q_h^{proc} supplies the upper bound on the battery's reserve commitment, while A_t^{up} and $Q_{h(t)}^{\text{proc}}$ produce the realized coefficient in Eq. (3). Gate-specific ECMWF weather runs are introduced in Section 9 below as forecast inputs rather than as ex-post observations. Energy, activation, and imbalance settlement use 15-minute intervals; aFRR capacity and the proxy inputs use hourly observations. Hourly quantities are mapped to their four settlement intervals without changing the UTC key. Equations (5) and (4) therefore feed the optimization branch, whereas Eq. (6) feeds the ex-post proxy branch.

5 System-level renewable energy utilization

For VRE technology $v \in \mathcal{V} = \{\text{wind}, \text{solar}\}$, let $P_{v,t}^{\text{avail}}$ denote the power that would be available without curtailment, and let $P_{v,t}^{\text{used}}$ denote the power accepted by the system after dispatch. The actual curtailed power is

$$P_{v,t}^{\text{curt}} = \left[P_{v,t}^{\text{avail}} - P_{v,t}^{\text{used}} \right]^+, \quad [x]^+ = \max(x, 0). \quad (7)$$

For an evaluation horizon $\mathcal{H} \subseteq \mathcal{T}$, curtailed energy, the curtailment rate, and the renewable utilization rate are

$$E_{\mathcal{H}}^{\text{curt}} = \sum_{t \in \mathcal{H}} \sum_{v \in \mathcal{V}} P_{v,t}^{\text{curt}} \Delta t, \quad (8)$$

$$CR_{\mathcal{H}} = \frac{E_{\mathcal{H}}^{\text{curt}}}{\sum_{t \in \mathcal{H}} \sum_{v \in \mathcal{V}} P_{v,t}^{\text{avail}} \Delta t}, \quad (9)$$

$$UR_{\mathcal{H}} = 1 - CR_{\mathcal{H}}. \quad (10)$$

Equations (7)–(10) require available and utilized generation to share the same technology, location, and temporal boundaries. Realized wind and solar generation already reflect dispatch, network constraints, and possible curtailment; they therefore cannot substitute for $P_{v,t}^{\text{avail}}$.

The contribution of battery storage to curtailment reduction requires a controlled system-level counterfactual. Let scenario 0 exclude the battery and scenario 1 include it, while holding demand, weather, available generation, network constraints, and all other generating-unit constraints fixed. The model-based curtailment reduction attributable to the battery is

$$\Delta E_{\text{BESS},\mathcal{H}}^{\text{curt}} = E_{\mathcal{H}}^{\text{curt},0} - E_{\mathcal{H}}^{\text{curt},1}. \quad (11)$$

A positive value in Eq. (11) can be reported as model-based curtailment reduction. A claim that the battery is charged from VRE additionally requires source tracing or an explicit coupling constraint, for example

$$0 \leq P_t^{\text{ch,VRE}} \leq \sum_{v \in \mathcal{V}} P_{v,t}^{\text{used}}, \quad P_t^{\text{ch,VRE}} \leq P_t^{\text{ch}}. \quad (12)$$

The current price-taking DK1 simulation does not provide the source tracing required by Eq. (12); hence, $P_t^{\text{ch,VRE}}$ is not evaluated.

6 DK1 observations and model-generated variables

The market, activation, production, consumption, and exchange observations come from the Energinet Energy Data Service datasets *DayAheadPrices*, *AfrrReservesNordic*, *ImbalancePrice*, and *ProductionConsumptionSettlement* [16]. Their original API metadata and responses are archived with retrieval timestamps and SHA-256 hashes. The 07:30 capacity gate and the publication of the capacity result before the 12:00 energy decision follow the Energinet DK1 aFRR capacity-market method and the Nordic implementation guide [17, 18]. Table 3 collects the symbols used throughout the paper, grouped by whether they are DK1 observations, archived forecasts, or model-generated decision and settlement outputs; it is a lookup reference rather than a set of new definitions.

Energinet hourly production and consumption fields are stored as MWh over one hour and are treated as the corresponding hourly average MW when broadcast to 15-minute slots; the numerical value is unchanged and multiplication by Δt produces interval MWh. The variables P^{ch} , P^{dis} , P^{rec} , r^{up} , and Q^{proc} are in MW; S , A^{up} , e^{req} , and e^{del} are in MWh; a is in MWh per MW. Energy, activation, and imbalance prices are in EUR/MWh, while the hourly capacity price is in EUR/MW.

Because of publication delays, $P_t^{\text{wind,obs}}$, $P_t^{\text{solar,obs}}$, and L_t^{obs} are used only for ex-post evaluation and are not inputs to the corresponding day-ahead decisions. Battery charging, discharging, and state of charge are outputs of the research model rather than observations from a physical DK1 battery.

7 Ex-post proxy metrics observable in DK1

Observed VRE power, the VRE-to-load ratio, the local residual-load proxy, and the local VRE-above-load proxy are defined as

$$P_t^{\text{VRE,obs}} = P_t^{\text{wind,obs}} + P_t^{\text{solar,obs}}, \quad (13)$$

$$\rho_t^{\text{VRE}} = \frac{P_t^{\text{VRE,obs}}}{L_t^{\text{obs}}}, \quad L_t^{\text{obs}} > 0, \quad (14)$$

$$NL_t^{\text{proxy}} = L_t^{\text{obs}} - P_t^{\text{VRE,obs}}, \quad (15)$$

$$X_t^{\text{proxy}} = [-NL_t^{\text{proxy}}]^+ = [P_t^{\text{VRE,obs}} - L_t^{\text{obs}}]^+. \quad (16)$$

Equation (13) first combines the two observed VRE technologies. Its output is divided by load in Eq. (14) and subtracted from load in Eq. (15). Taking the positive part of the negative residual

Table 3: Variables used in the utilization definitions and DK1 proxy metrics.

Symbol	Data role	Definition
$P_t^{\text{wind,obs}}$	DK1 ex-post observation	Realized wind generation
$P_t^{\text{solar,obs}}$	DK1 ex-post observation	Realized solar generation
L_t^{obs}	DK1 ex-post observation	Total load
$F_{b,t}$	DK1 ex-post observation	Signed exchange on border b ; import positive
$C_{k,t}$	DK1 ex-post observation	Generation in non-VRE category k
H_t	DK1 ex-post observation	Power-to-Heat consumption included in total load
$w_{d,h}^{(g)}$	Archived forecast	Weather vector visible at gate g
$\hat{L}_{d,h}^{(g)}$	Model-generated	Gate-specific load forecast
λ_t^{DA}	DK1 market observation	Day-ahead energy price
κ_h^{up}	DK1 market observation	Upward-aFRR capacity price
Q_h^{proc}	DK1 market observation	Total procured upward-aFRR capacity
A_t^{up}	DK1 settlement observation	Activated upward energy
a_t	Derived settlement input	Activation energy per MW of procured reserve
λ_t^{act}	DK1 settlement observation	Upward activation price
λ_t^{imb}	DK1 settlement observation	Imbalance and recovery price
P_t^{ch}	Model-generated	Battery charging power
P_t^{dis}	Model-generated	Battery discharging power
r_h^{up}	Model-generated	Hourly upward-aFRR commitment
e_t^{del}	Model-generated	Delivered activation energy
P_t^{rec}	Model-generated	Paid SOC-recovery power
S_t^{plan}	Model-generated	Pre-activation planned state of charge
S_t^{real}	Model-generated	State after activation and recovery
$P_{v,t}^{\text{avail}}$	Unobserved	Unconstrained VRE availability
$P_{v,t}^{\text{curt}}$	Unobserved	Actual curtailed VRE power

load then produces X^{proxy} in Eq. (16). This output is reused in the balance decomposition, the charging-overlap metrics, the forecast target, and the positive-hour classification; none of those downstream calculations changes its definition. Neither NL_t^{proxy} nor X_t^{proxy} accounts for conventional generation, cross-border exchanges, network constraints, or actual curtailment. They must therefore be described as a local residual-load proxy and a local VRE-above-load proxy, respectively, rather than as system net load or actual renewable surplus.

7.1 Exchange, conventional-generation, and Power-to-Heat decomposition

The signed net exchange, gross export, gross import, and observed non-VRE generation are

$$F_t = \sum_{b \in \mathcal{B}} F_{b,t}, \quad E_t = [-F_t]^+, \quad I_t = [F_t]^+, \quad (17)$$

$$C_t = \sum_{k \in \mathcal{K}} C_{k,t}, \quad \mathcal{K} = \{\text{central, local, commercial, local self-consumption, hydro, unknown}\}. \quad (18)$$

Equations (17) and (18) aggregate the border and generation-category inputs in Eq. (6); their outputs enter the measured balance below. The Energinet settlement table obeys the hourly balance

$$P_t^{\text{VRE,obs}} + C_t + F_t = L_t^{\text{obs}} + \varepsilon_t, \quad (19)$$

where ε_t is the numerical balance residual. Substituting Eq. (19) into Eq. (16) proceeds through

$$P_t^{\text{VRE,obs}} - L_t^{\text{obs}} = -C_t - F_t + \varepsilon_t = E_t - I_t - C_t + \varepsilon_t, \quad (20)$$

because $F_t = I_t - E_t$. Applying the same positive-part operator used in Eq. (16) gives

$$X_t^{\text{proxy}} = [E_t - I_t - C_t + \varepsilon_t]^+. \quad (21)$$

Thus a proxy-positive hour may be an exporting hour rather than an hour in which energy lacks a system use. The export coverage and the conventional share of export over a set $\mathcal{H}$ are reported without adding their numerators:

$$EC_{\mathcal{H}} = \frac{\sum_{t \in \mathcal{H}} \min(X_t^{\text{proxy}}, E_t)}{\sum_{t \in \mathcal{H}} X_t^{\text{proxy}}}, \quad (22)$$

$$CS_{\mathcal{H}} = \frac{\sum_{t \in \mathcal{H}} C_t}{\sum_{t \in \mathcal{H}} E_t}. \quad (23)$$

Power-to-Heat is already contained in L_t^{obs} . It is therefore not subtracted again in Eq. (21). Its observed absorption is evaluated through the explicit no-Power-to-Heat counterfactual

$$X_t^{\text{noP2H}} = \left[P_t^{\text{VRE,obs}} - (L_t^{\text{obs}} - H_t) \right]^+, \quad (24)$$

$$A_t^{\text{P2H}} = X_t^{\text{noP2H}} - X_t^{\text{proxy}}, \quad (25)$$

$$HS_{\mathcal{H}} = \frac{\sum_{t \in \mathcal{H}} A_t^{\text{P2H}}}{\sum_{t \in \mathcal{H}} X_t^{\text{noP2H}}}. \quad (26)$$

A_t^{P2H} is an observed accounting counterfactual, not an estimate of curtailment avoided by electric heating. The dependency is $H_t \rightarrow (L_t^{\text{obs}} - H_t) \rightarrow X_t^{\text{noP2H}} \rightarrow A_t^{\text{P2H}} \rightarrow HS_{\mathcal{H}}$; H_t is not an additional subtraction from Eq. (21).

The simulated midday charging energy on delivery day d is

$$E_{\text{ch},d}^{\text{mid}} = \sum_{t \in \mathcal{M}_d} P_t^{\text{ch}} \Delta t. \quad (27)$$

The temporal overlap between simulated charging and the local VRE-above-load proxy is

$$E_{\text{ch},d}^{\text{overlap}} = \sum_{t \in \mathcal{M}_d} \min(P_t^{\text{ch}}, X_t^{\text{proxy}}) \Delta t. \quad (28)$$

Two proxy ratios with distinct denominators are then reported:

$$AS_d^{\text{ch}} = \frac{E_{\text{ch},d}^{\text{overlap}}}{E_{\text{ch},d}^{\text{mid}}}, \quad E_{\text{ch},d}^{\text{mid}} > 0, \quad (29)$$

$$AR_d^{\text{proxy}} = \frac{E_{\text{ch},d}^{\text{overlap}}}{\sum_{t \in \mathcal{M}_d} X_t^{\text{proxy}} \Delta t}, \quad \sum_{t \in \mathcal{M}_d} X_t^{\text{proxy}} \Delta t > 0. \quad (30)$$

AS_d^{ch} is the share of simulated midday charging that coincides with the local VRE-above-load proxy. AR_d^{proxy} is the share of that proxy whose power magnitude temporally overlaps with simulated charging. Both are temporal alignment metrics. Neither traces the physical origin of electricity nor equals the renewable utilization rate in Eq. (10). Both ratios inherit the same numerator from Eq. (28); only their denominators differ. The numerator couples the deployed charge from Eq. (52) with the ex-post proxy from Eq. (16), so it cannot be computed before both the decision and realized-observation branches are available.

To avoid relying exclusively on the strict condition $P_t^{\text{VRE,obs}} > L_t^{\text{obs}}$, a renewable-rich indicator may be defined using a threshold τ . The threshold must be fixed using the training period or an independent reference period and must not be selected from test outcomes:

$$I_t^{\text{rich}} = \mathbb{I}(\rho_t^{\text{VRE}} \geq \tau), \quad (31)$$

$$E_{\text{ch},d}^{\text{rich}} = \sum_{t \in \mathcal{M}_d} I_t^{\text{rich}} P_t^{\text{ch}} \Delta t. \quad (32)$$

Until τ is frozen on an independent sample, Eq. (32) is a sensitivity metric rather than a confirmatory primary outcome.

Charging during negative day-ahead prices is defined as

$$E_{\text{ch},d}^{\text{neg}} = \sum_{t \in \mathcal{M}_d} \mathbb{I}(\lambda_t^{\text{DA}} < 0) P_t^{\text{ch}} \Delta t. \quad (33)$$

Equation (33) measures an economic response to negative prices. Negative prices can indicate insufficient system flexibility, but they do not independently establish that wind or solar generation was curtailed.

8 Market forecasting and scenario construction

The market-forecast branch is evaluated separately from the WLPF branch. It produces the four-target vector $\widehat{\Xi}_{d,s}^{(g)}$ in Eq. (4). The point baselines are previous-day persistence (PERS) [1], the Lasso-estimated autoregressive model (LEAR) [2], and LightGBM (LGBM) [3]. These methods are established comparators, not contributions of this paper. The frozen LGBM arm supplies point predictions for all four targets; the separately trained LEAR and LGBM reference arms are available for day-ahead and capacity prices.

All probabilistic models use a rolling origin and the same chronological information blocks. For delivery day d , the marginal fit window is the 84 calendar days $d - 119$ through $d - 36$, probability calibration uses $d - 35$ through $d - 8$, and $d - 7$ through $d - 1$ is a clearing gap. No missing target or feature is filled. Quantile levels are

$$\mathcal{A} = \{0.005, 0.01, 0.025, 0.05, 0.1, 0.2, \dots, 0.8, 0.9, 0.95, 0.975, 0.99, 0.995\}. \quad (34)$$

For day-ahead price, capacity price, and activation price, conditional quantiles are estimated by linear quantile regression [4]. Day-ahead and capacity models are hour specific; activation price is pooled over the 24 hours with six four-hour period indicators. Predictors contain only gate-visible lagged values, seven-day visible means, and calendar terms. The activation-volume marginal is a hurdle model: a logistic regression estimates the zero-event probability and quantile regression on log positive volumes estimates the conditional positive distribution. Estimated quantiles are sorted to remove crossing and linearly interpolated when a scenario requests a probability between grid points.

GCJS is the joint scenario model proposed and implemented in this paper. Let $u_{d,j}$ be the randomized probability integral transform (PIT) of dimension j , where j indexes the $4 \times 24 = 96$ target-hour cells, and let

$$z_{d,j} = \Phi^{-1}(\min\{0.999, \max(0.001, u_{d,j})\}). \quad (35)$$

Pairwise-complete correlations of $z_{d,j}$ form R_d . A principal-axis factor model gives the regularized Gaussian-copula correlation

$$\Sigma_d = B_d B_d^\top + \Psi_d, \quad \text{diag}(\Sigma_d) = \mathbf{1}, \quad (36)$$

where Ψ_d is diagonal. Ranks one, two, and three are compared only on the probability-calibration block, and the rank with the lowest standardized joint Energy Score is selected. Drawing $Z_s \sim \mathcal{N}(0, \Sigma_d)$, applying Φ componentwise, and mapping the resulting uniforms through the marginal inverse quantiles yields 100 forecast scenarios; 50 equal-weight scenarios are retained for optimization. This is a Gaussian-copula construction in the standard sense of Nelsen [5], while the rolling marginals, gate-safe information sets, low-rank selection, and 96-dimensional DK1 target organization are the present implementation.

The scenario comparisons isolate distinct modeling choices. Rolling climatology and persistence are classical reference forecasts [1]. Independent quantile regression (QR-Ind) samples every marginal independently. Historical paired scenarios (Hist-P) resample one complete historical day for all targets, whereas historical independent scenarios (Hist-I) resample a day separately for each target; these are present implementations of empirical sample-based scenarios [11]. The empirical-copula comparator (EC) resamples complete PIT-rank rows. PGC-NoCross sets cross-target correlations to zero while retaining within-target time dependence; PGC-NoTime sets cross-hour correlations to zero while retaining same-hour cross-target dependence. These two arms and QR-Ind are the formal dependence ablations. The sequential block-copula comparator implements the multi-phase dependence structure of Al-Lawati et al. [14] under the common DK1 marginals. The weather-augmented arm (WX-PGC) is an explicit information sensitivity using published wind and solar forecasts; it is not part of GCJS and is evaluated on the 209 days for which those inputs are complete.

At the 12:00 gate, the already published capacity-price path may condition the remaining 72 market dimensions. GCJS uses the Gaussian conditional mean and Schur-complement covariance; EC uses the ten nearest historical capacity paths. These are information-update experiments, not separate proposed methods. Their empirical performance is reported before any decision result, because a later gate does not by itself guarantee a better forecast.

Point errors use MAE and RMSE. Marginal probabilistic performance uses the pinball loss, sample CRPS, PIT distance, interval coverage, and the activation zero-event Brier score. Joint scenarios use the Energy Score and Variogram Score [6]. Daily loss differences are tested with the Diebold–Mariano statistic [7] and six-lag Newey–West covariance [8]; uncertainty intervals use a seven-day moving-block bootstrap [9] with 2,000 replications. All forecast scores are negatively oriented: lower is better.

9 Gate-available weather, load, and proxy forecasts: WLPF

WLPF is proposed in this paper as the gate-safe renewable-alignment signal for GRASCO. It is distinct from WX-PGC: WX-PGC augments the market marginals, whereas WLPF predicts the ex-post VRE-above-load proxy in Eq. (16).

For each delivery day d , archived ECMWF IFS forecasts are selected by model initialization time rather than by the data-download time. Let $r(d, g)$ denote the selected run and impose a conservative six-hour production delay:

$$r(d, 07:30) = (d - 2) \text{ 18:00 UTC}, \quad a(d, 07:30) = r(d, 07:30) + 6 \text{ h}, \quad (37)$$

$$r(d, 12:00) = (d - 1) \text{ 00:00 UTC}, \quad a(d, 12:00) = r(d, 12:00) + 6 \text{ h}. \quad (38)$$

The implementation asserts $a(d, g) \leq G(d, g)$, where $G(d, g)$ is the UTC instant corresponding to the civil-time gate on day $d - 1$. The weather vector $w_{d,h}^{(g)}$ contains the equal-grid DK1 means of 2-m temperature, 100-m wind speed, shortwave radiation, and total cloud cover, together with the spatial standard deviation of 100-m wind speed, from run $r(d, g)$. The run archive and its distinction between initialization time and public availability are documented by Open-Meteo [23]. The underlying numerical weather prediction system is ECMWF IFS HRES [19]; the retrieved fields are archived model forecasts rather than weather observations.

Weather fields are sampled on a 3×3 grid defined by latitudes $\{55.5, 56.25, 57.0\}^\circ\text{N}$ and longitudes $\{8.0, 9.0, 10.0\}^\circ\text{E}$, yielding nine points over Jutland and adjacent waters. For each valid hour and model run, the weather variables are aggregated using an unweighted arithmetic mean across the nine points. For a weather field x and grid index j , the aggregation is

$$\bar{x}_{d,h}^{(g)} = \frac{1}{9} \sum_{j=1}^9 x_{d,h,j}^{(g)}, \quad (39)$$

$$\sigma_{U,d,h}^{(g)} = \left[\frac{1}{9} \sum_{j=1}^9 \left(U_{d,h,j}^{(g)} - \bar{U}_{d,h}^{(g)} \right)^2 \right]^{1/2}. \quad (40)$$

The weather vector is therefore

$$w_{d,h}^{(g)} = \left[\bar{T}_{d,h}^{(g)}, \bar{U}_{d,h}^{(g)}, \sigma_{U,d,h}^{(g)}, \bar{R}_{d,h}^{(g)}, \bar{C}_{d,h}^{(g)} \right]^\top, \quad (41)$$

where T, U, R, C denote 2-m temperature, 100-m wind speed, shortwave radiation, and total cloud cover. No bidding-zone boundary mask, land-sea mask, area weighting, or installed-capacity weighting is applied. These fields should therefore be interpreted as regional DK1 weather proxies rather than exact spatial averages over the DK1 bidding zone.

Separate component models are fitted for $q \in \{\text{wind, solar, load}\}$. The design vector is

$$\phi_{q,d,h}^{(g)} = \left[1, \text{Fourier}_4(h), \text{DoW}(d), w_{d,h}^{(g)}, T_{d,h}^2, U_{d,h}^2, U_{d,h}^3, q_{d-7,h}^{\text{obs}}, q_{d-14,h}^{\text{obs}} \right]^\top, \quad (42)$$

where T and U are temperature and 100-m wind speed, respectively; $\text{Fourier}_4(h)$ contains sine and cosine terms for harmonics one through four; and $\text{DoW}(d)$ contains six weekday indicators with Monday omitted. Ordinary least squares is fitted only on the existing 84-calendar-day training window from $d - 119$ through $d - 36$. Define

$$\mathcal{F}_d = \{(d', h) : d - 119 \leq d' \leq d - 36 \text{ and every required input and target is observed}\}. \quad (43)$$

The fitted coefficients and nonnegative component forecast are then

$$\hat{\beta}_{q,d}^{(g)} = \arg \min_{\beta} \sum_{(d', h) \in \mathcal{F}_d} \left(q_{d',h}^{\text{obs}} - \phi_{q,d',h}^{(g)\top} \beta \right)^2, \quad (44)$$

$$\hat{q}_{d,h}^{(g)} = \left[\phi_{q,d,h}^{(g)\top} \hat{\beta}_{q,d}^{(g)} \right]^+. \quad (45)$$

The continuous dependency is $r(d, g) \rightarrow w_{d,h}^{(g)} \rightarrow \phi_{q,d,h}^{(g)} \rightarrow \hat{\beta}_{q,d}^{(g)} \rightarrow \hat{q}_{d,h}^{(g)}$. No target from delivery day d enters $\mathcal{F}_d$. Missing inputs remain missing and the affected forecast is not evaluated. The observed 7- and 14-day lags are assigned the same 75.75-hour publication delay used by the DK1 data audit, and the implementation separately verifies their availability at each gate. The gate-available proxy forecast is then

$$\hat{X}_{d,h}^{\text{proxy},g} = \left[\hat{P}_{d,h}^{\text{wind},g} + \hat{P}_{d,h}^{\text{solar},g} - \hat{L}_{d,h}^g \right]^+. \quad (46)$$

Thus Eqs. (43)–(45) produce three component forecasts, and Eq. (46) combines their outputs using the same positive-part operation as the realized target in Eq. (16). Equation (46) is a model-generated forecast of the same ex-post proxy in Eq. (16); it is neither an official TSO load forecast nor a forecast of physical curtailment. It is evaluated as a forecast target and supplied only to the explicitly labelled proxy-aware arms in Eq. (72); the market-only arms retain the original information set and provide the control comparison.

For every gate and evaluation set $\mathcal{Q}$, the implementation reports component MAE, proxy MAE and RMSE, and positive-hour precision and recall:

$$\text{MAE}_q^g = \frac{1}{|\mathcal{Q}|} \sum_{(d,h) \in \mathcal{Q}} \left| \hat{q}_{d,h}^g - q_{d,h}^{\text{obs}} \right|, \quad (47)$$

$$\text{MAE}_X^g = \frac{1}{|\mathcal{Q}|} \sum_{(d,h) \in \mathcal{Q}} \left| \hat{X}_{d,h}^{\text{proxy},g} - X_{d,h}^{\text{proxy}} \right|, \quad (48)$$

$$\text{RMSE}_X^g = \sqrt{\frac{1}{|\mathcal{Q}|} \sum_{(d,h) \in \mathcal{Q}} \left(\hat{X}_{d,h}^{\text{proxy},g} - X_{d,h}^{\text{proxy}} \right)^2}, \quad (49)$$

$$\text{Precision}_X^g = \frac{\sum_{(d,h) \in \mathcal{Q}} \hat{I}_{d,h}^g I_{d,h}}{\sum_{(d,h) \in \mathcal{Q}} \hat{I}_{d,h}^g}, \quad \text{Recall}_X^g = \frac{\sum_{(d,h) \in \mathcal{Q}} \hat{I}_{d,h}^g I_{d,h}}{\sum_{(d,h) \in \mathcal{Q}} I_{d,h}}, \quad (50)$$

where $I_{d,h} = \mathbb{I}(X_{d,h}^{\text{proxy}} > 0)$ and $\hat{I}_{d,h}^g = \mathbb{I}(\hat{X}_{d,h}^{\text{proxy},g} > 0)$. Equations (47)–(50) compare the forecast branch with its realized counterpart; they do not feed any decision variable. The information value between the gates is summarized by

$$\text{MAR}_{07:30 \rightarrow 12:00} = \frac{1}{|\mathcal{Q}|} \sum_{(d,h) \in \mathcal{Q}} \left| \hat{X}_{d,h}^{\text{proxy},12:00} - \hat{X}_{d,h}^{\text{proxy},07:30} \right|. \quad (51)$$

10 Proposed GRASCO decision model

GRASCO is proposed in this paper. Its two-gate nonanticipativity follows the actual DK1 information sequence, while its stochastic-programming ingredients are related to established chance-constrained, sequential, and three-stage storage formulations [13, 14, 15]. The new element evaluated here is the explicit coupling of a gate-safe renewable-rich forecast to the two-gate energy-reserve schedule and its subsequent 15-minute realized settlement trace.

The common battery has rated power $\bar{P} = 4$ MW, energy capacity $\bar{S} = 8$ MWh, round-trip efficiency $\eta^{\text{rt}} = 0.85$, one-way efficiencies $\eta_c = \eta_d = \sqrt{0.85}$, and initial and terminal SOC $S_0 = S_T = 4.4$ MWh. The market-share parameter is $\alpha^{\text{share}} = 1$ for the main arm. The programme contains three variable layers shown in Table 4.

At 07:30, the capacity-price paths are standardized by their historical hour-specific standard deviations and assigned to clusters $c(s)$. Energy variables share one value within a cluster, which is the implemented nonanticipativity rule. These 07:30 cluster energy plans value the common reserve decision but are not deployed. At 12:00 the realized capacity-price path is inserted, all

Table 4: Decision layers and their information dependence.

Layer	Variables	Dependence
Common reserve	$r_{d,h}^{\text{up}}$	identical in every scenario
Cluster energy plan	$P_{d,c,t}^{\text{ch}}, P_{d,c,t}^{\text{dis}}, S_{d,c,t}^{\text{plan}}$	only on $c = c(s)$
Scenario recourse	$P_{d,s,t}^{\text{rec}}, e_{d,s,t}^{\text{del}}, S_{d,s,t}^{\text{real}}$	full scenario s

scenario labels are set to one cluster, the 07:30 reserve vector is fixed, and the resulting unique energy plan is deployed:

$$u_d = \{r_{d,h}^{\text{up}}, P_{d,t}^{\text{ch}}, P_{d,t}^{\text{dis}}, S_{d,t}^{\text{plan}}\}. \quad (52)$$

The complete stochastic-programme vector, including variables used only inside scenario evaluation, is

$$x_d = \{r_{d,h}^{\text{up}}, P_{d,c,t}^{\text{ch}}, P_{d,c,t}^{\text{dis}}, S_{d,c,t}^{\text{plan}}, P_{d,s,t}^{\text{rec}}, e_{d,s,t}^{\text{del}}, S_{d,s,t}^{\text{real}}, q_{d,c,h}^g\}. \quad (53)$$

Thus $u_d \subset x_d$: x_d is optimized, whereas u_d is the frozen plan passed to ex-post settlement.

The procured market volume and battery rating bound the common reserve before any SOC equation is evaluated:

$$0 \leq r_{d,h}^{\text{up}} \leq \min(\bar{P}, \alpha^{\text{share}} Q_{d,h}^{\text{proc}}). \quad (54)$$

Thus Q^{proc} is an upstream feasibility input, not a revenue or activation variable.

For each capacity-price cluster, the pre-activation plan starts from S_0 and follows

$$S_{d,c,t}^{\text{plan}} = S_{d,c,t-1}^{\text{plan}} + \eta_c \Delta t P_{d,c,t}^{\text{ch}} - \frac{\Delta t}{\eta_d} P_{d,c,t}^{\text{dis}}, \quad S_{d,c,0}^{\text{plan}} = S_0. \quad (55)$$

Its state, terminal, power, and one-hour reserve-energy constraints are

$$0 \leq S_{d,c,t}^{\text{plan}} \leq \bar{S}, \quad S_{d,c,T}^{\text{plan}} = S_0, \quad (56)$$

$$P_{d,c,t}^{\text{ch}} + P_{d,c,t}^{\text{dis}} + r_{d,h(t)}^{\text{up}} \leq \bar{P}, \quad (57)$$

$$S_{d,c,t}^{\text{plan}} \geq \frac{T^{\text{res}}}{\eta_d} r_{d,h(t)}^{\text{up}}, \quad T^{\text{res}} = 1 \text{ h}. \quad (58)$$

Equation (55) supplies the state used by Eqs. (56)–(58); the reserve cannot be checked without first propagating the charge and discharge plan.

For scenario s , the cluster plan $c(s)$ and the activation coefficient from the market forecast vector produce requested activation energy:

$$e_{d,s,t}^{\text{req}} = r_{d,h(t)}^{\text{up}} a_{d,s,t}. \quad (59)$$

Delivered activation and paid recovery then modify the planned trajectory. The scenario state is

$$S_{d,s,t}^{\text{real}} = S_{d,s,t-1}^{\text{real}} + \eta_c \Delta t (P_{d,c(s),t}^{\text{ch}} + P_{d,s,t}^{\text{rec}}) - \frac{\Delta t}{\eta_d} P_{d,c(s),t}^{\text{dis}} - \frac{e_{d,s,t}^{\text{del}}}{\eta_d},$$

$$S_{d,s,0}^{\text{real}} = S_0. \quad (60)$$

The recourse variables are coupled by

$$0 \leq S_{d,s,t}^{\text{real}} \leq \bar{S}, \quad S_{d,s,T}^{\text{real}} = S_0, \quad (61)$$

$$P_{d,c(s),t}^{\text{ch}} + P_{d,c(s),t}^{\text{dis}} + P_{d,s,t}^{\text{rec}} + \frac{e_{d,s,t}^{\text{del}}}{\Delta t} \leq \bar{P}, \quad (62)$$

$$S_{d,s,t}^{\text{real}} + \frac{e_{d,s,t}^{\text{del}}}{\eta_d} \geq \frac{T^{\text{res}}}{\eta_d} r_{d,h(t)}^{\text{up}}, \quad (63)$$

$$0 \leq e_{d,s,t}^{\text{del}} \leq e_{d,s,t}^{\text{req}}. \quad (64)$$

The configured 07:30 “full” delivery mode replaces Eq. (64) by $e_{d,s,t}^{\text{del}} = e_{d,s,t}^{\text{req}}$ inside the early reserve valuation. The 12:00 “delivery-first” mode retains the inequality and uses the lexicographic solve below. Equation (62) tests power headroom after the energy plan, recovery, and activation are combined. Equation (63) adds the current delivered energy back to the post-interval state and checks that the pre-delivery state covered the one-hour reserve obligation. These two constraints separate power-driven shortfall from energy-driven shortfall.

The scenario settlement profit is

$$\begin{aligned}\Pi_{d,s}(x_d) = & \sum_t \Delta t \lambda_{d,s,t}^{\text{DA}} \left(P_{d,c(s),t}^{\text{dis}} - P_{d,c(s),t}^{\text{ch}} \right) + \sum_h \kappa_{d,s,h}^{\text{up}} r_{d,h}^{\text{up}} \\ & + \sum_t \lambda_{d,s,t}^{\text{act}} e_{d,s,t}^{\text{del}} - \sum_t \lambda_{d,s,t}^{\text{imb}} \left(e_{d,s,t}^{\text{req}} - e_{d,s,t}^{\text{del}} \right) \\ & - \sum_t \Delta t \lambda_{d,s,t}^{\text{imb}} P_{d,s,t}^{\text{rec}}.\end{aligned}\quad (65)$$

The five terms in Eq. (65) are day-ahead energy value, capacity revenue, delivered-activation revenue, undelivered-activation cost, and recovery-energy cost. Every monetary term therefore depends on a physical quantity produced by Eqs. (54)–(64).

For lower-tail profit risk, introduce a threshold ζ and nonnegative shortfalls ξ_s . The implemented lower-tail CVaR representation follows Rockafellar and Uryasev [10]:

$$\text{CVaR}_\beta^{\text{lower}}(\Pi) = \max_{\zeta, \xi} \left\{ \zeta - \frac{1}{1-\beta} \sum_s \pi_s \xi_s \right\}, \quad (66)$$

$$\xi_s \geq \zeta - \Pi_{d,s}, \quad \xi_s \geq 0. \quad (67)$$

With risk weight χ and $\beta = 0.95$, the stochastic objective is

$$\max_{x_d, \zeta, \xi} (1-\chi) \sum_{s \in \mathcal{S}_d} \pi_s \Pi_{d,s}(x_d) + \chi \text{CVaR}_\beta^{\text{lower}}(\Pi_{d,s}(x_d)). \quad (68)$$

where Eqs. (66)–(67) supply the second term. When $\chi = 0$, only expected profit remains; when $\chi > 0$, low-profit scenarios affect the solution through ξ_s .

The GRASCO sensitivity arms add a gate-visible charge-alignment quantity without changing the market settlement ledger. Let $\omega_{d,c} = \sum_{s:c(s)=c} \pi_s$ be the probability mass assigned to capacity-price cluster c . For gate g , cluster c , and an hour in the fixed midday set $\mathcal{M}_d$, define

$$0 \leq q_{d,c,h}^g \leq \hat{X}_{d,h}^{\text{proxy},g}, \quad (69)$$

$$q_{d,c,h}^g \leq \Delta t \sum_{t:h(t)=h} P_{d,c,t}^{\text{ch}}. \quad (70)$$

Thus $q_{d,c,h}^g$ is the portion of planned hourly charging that can be matched to the forecast proxy, rather than an assertion about the physical source of the charged electricity. Expected forecast-aligned charging is

$$A_d^g(x_d) = \sum_c \omega_{d,c} \sum_{h \in \mathcal{M}_d} q_{d,c,h}^g. \quad (71)$$

Writing the market-only objective in Eq. (68) as $J_d^{\text{market}}(x_d)$, the exploratory family solves

$$\max_{x_d, \zeta, \xi} J_d^{\text{market}}(x_d) + \rho A_d^g(x_d), \quad \rho \in \{0, 25, 50, 100\} \text{ EUR/MWh}. \quad (72)$$

The coefficient ρ is an experimental alignment credit, not observed revenue. The $\rho = 0$ arm is the market-only control, and the positive values form a sensitivity comparison rather than a

selected social value. CVaR is still calculated from $\Pi_{d,s}$ alone, and all reported realized profit excludes ρA_d^g .

At 07:30 the common reserve is optimized with the configured delivery mode. At 12:00 that reserve is fixed, the observed capacity price replaces its scenario forecast, and the delivery-first implementation is lexicographic. It first obtains

$$D_d^* = \max_s \sum_s \pi_s \sum_t e_{d,s,t}^{\text{del}}, \quad (73)$$

and then solves Eq. (72) subject to expected delivery of at least $D_d^* - \varepsilon_D$, where $\varepsilon_D = 10^{-9}$ MWh. At 07:30 Eq. (72) uses $\hat{X}^{\text{proxy},07:30}$; at 12:00 the frozen reserve vector is combined with $\hat{X}^{\text{proxy},12:00}$. This prevents the noon update from exchanging material delivery capability for a small market or alignment improvement. The output of this second solve is the deployed vector in Eq. (52); Eq. (73) supplies its binding delivery floor, and realized market data then carry that vector into the settlement trace below.

11 Fifteen-minute settlement trace and linkage diagnostics

For each deployed strategy, realized activation first converts the frozen reserve decision into an observed request:

$$e_{d,t}^{\text{req,obs}} = r_{d,h(t)}^{\text{up}} a_{d,t}^{\text{obs}}. \quad (74)$$

Holding u_d fixed, the settlement closure maximizes total delivered activation subject to Eqs. (60)–(64) with realized inputs, then minimizes the imbalance-price cost of P^{rec} . This closure produces e^{del} , P^{rec} , and S^{real} ; it does not change charge, discharge, or reserve. The ex-post trace records

$$z_{d,t} = \left(P_t^{\text{ch}}, P_t^{\text{dis}}, r_{h(t)}^{\text{up}}, e_t^{\text{req}}, e_t^{\text{del}}, P_t^{\text{rec}}, S_t^{\text{plan}}, S_t^{\text{real}}, \Pi_t^{\text{energy}}, \Pi_t^{\text{cap}}, \Pi_t^{\text{act}}, C_t^{\text{short}}, C_t^{\text{rec}} \right). \quad (75)$$

Let $n_{d,h} = |\{t \in \mathcal{T}_d : h(t) = h\}|$. The trace components are linked to the physical variables by

$$\Pi_{d,t}^{\text{energy}} = \Delta t \lambda_{d,t}^{\text{DA}} \left(P_{d,t}^{\text{dis}} - P_{d,t}^{\text{ch}} \right), \quad (76)$$

$$\Pi_{d,t}^{\text{cap}} = \frac{\kappa_{d,h(t)}^{\text{up}} r_{d,h(t)}^{\text{up}}}{n_{d,h(t)}}, \quad (77)$$

$$\Pi_{d,t}^{\text{act}} = \lambda_{d,t}^{\text{act}} e_{d,t}^{\text{del}}, \quad (78)$$

$$C_{d,t}^{\text{short}} = \lambda_{d,t}^{\text{imb}} \left(e_{d,t}^{\text{req,obs}} - e_{d,t}^{\text{del}} \right), \quad (79)$$

$$C_{d,t}^{\text{rec}} = \Delta t \lambda_{d,t}^{\text{imb}} P_{d,t}^{\text{rec}}. \quad (80)$$

The variables listed in Eq. (75) are therefore either frozen physical decisions, realized closure outputs, or the five monetary components in Eqs. (76)–(80). Hourly capacity revenue is therefore allocated equally across its observed 15-minute slots. Summing $\Pi^{\text{energy}} + \Pi^{\text{cap}} + \Pi^{\text{act}} - C^{\text{short}} - C^{\text{rec}}$ over t reproduces the daily settlement profit Π_d . This equality is the trace-to-daily audit used below.

The daily diagnostic vector is

$$J_d = \left(E_{\text{ch},d}^{\text{mid}}, A_d^{12:00}, E_{\text{ch},d}^{\text{overlap}}, AS_d^{\text{ch}}, AR_d^{\text{proxy}}, S_{d,17}, R_d^{\text{eve}}, E_d^{\text{req,eve}}, E_d^{\text{del,eve}}, U_d^{\text{eve}}, C_d^{\text{rec}}, \Pi_d \right), \quad (81)$$

The first two entries record midday charge and its overlap with the 12:00 forecast; the next three couple the ex-post proxy branch to deployed charge; $S_{d,17}$ is read from the realized SOC trace; the next four entries aggregate the frozen reserve and realized activation over $\mathcal{E}_d$; and the final

two entries aggregate the monetary trace. Thus J_d is downstream of both X^{proxy} and $z_{d,t}$, as shown in Eqs. (C1) and (C3). where

$$R_d^{\text{eve}} = \sum_{t \in \mathcal{E}_d} r_{h(t)}^{\text{up}} \Delta t, \quad (82)$$

$$E_d^{\text{req,eve}} = \sum_{t \in \mathcal{E}_d} e_t^{\text{req}}, \quad E_d^{\text{del,eve}} = \sum_{t \in \mathcal{E}_d} e_t^{\text{del}}, \quad (83)$$

$$U_d^{\text{eve}} = E_d^{\text{req,eve}} - E_d^{\text{del,eve}}, \quad DR^{\text{eve}} = \frac{\sum_d E_d^{\text{del,eve}}}{\sum_d E_d^{\text{req,eve}}}, \quad (84)$$

$$C_d^{\text{rec}} = \sum_t \Delta t \lambda_t^{\text{imb}} P_t^{\text{rec}}. \quad (85)$$

Equations (82)–(84) aggregate the hourly reserve and interval activation entries of Eq. (81); Eq. (85) is the daily sum of Eq. (80). The observable linkage is summarized by

$$\Gamma_S = \text{Corr}\left(E_{\text{ch},d}^{\text{mid}}, S_{d,17}\right), \quad (86)$$

$$\Gamma_R = \text{Corr}\left(E_{\text{ch},d}^{\text{mid}}, R_d^{\text{eve}}\right). \quad (87)$$

Equations (86) and (87) reuse the same upstream variable $E_{\text{ch},d}^{\text{mid}}$ but connect it to two different downstream outcomes: stored energy and reserve commitment. A physically useful midday–evening link requires $\Gamma_S > 0$. A decision protocol that actively converts midday charging into additional evening reserve would also be expected to have $\Gamma_R \geq 0$, unless the reserve decision is already power-saturated or frozen before the midday information becomes available.

12 Experimental design and method provenance

Table 5 fixes the names and evidence roles used in all result tables. Internal experiment identifiers are deliberately omitted: they are implementation keys, not scientific method names.

Accordingly, every uncited row in Table 5 is explicitly marked as proposed in this paper; present constructions, ablations, and sensitivities cite the established method on which they operate.

In decision tables, the suffix DO denotes deterministic optimization driven by a point forecast, and SO denotes stochastic optimization driven by scenarios. Thus LGBM-DO, Hist-P-SO, GCJS-SO, and PGC-SO identify the forecasting input and the common downstream optimizer without introducing additional methods.

DD-CC, MPF, and W&SPP are reconstructed under the same DK1 data, battery, feasible set, and settlement ledger as GRASCO. DD-CC transfers the activation distribution and chance-constraint idea of Toubreau et al. [13], but replaces the Belgian bilevel market and joint delivery guarantee by the frozen DK1 hourly quantile approximation. MPF transfers the sequential scenario-update structure of Al-Lawati et al. [14], but reverses the original day-ahead-to-reserve order to the DK1 reserve-to-day-ahead order, uses 50 rather than 90 scenarios, and removes the wind farm, intraday market, and down reserve. W&SPP transfers the three-stage scenario-tree and expectation–CVaR structure of Mancini et al. [15], but replaces the German four-hour product and 1.5-hour reserve duration by hourly DK1 aFRR and the common one-hour assumption, removes the co-located wind farm and binaries, and uses task-specified scenario counts and risk levels. These differences preclude claims of numerical replication; the comparison tests the published decision ideas under one controlled DK1 protocol. Table 6 fixes every numerical choice shared by all methods above, so that the Results section below can report differences without repeating these settings.

All numerical results below are authors’ calculations from archived Energinet and ECMWF data [16, 23]. Under `p1_paper/results`, forecast outputs are stored in `s3_forecast_side`,

Table 5: Method provenance and experimental role.

Short name	Method	Provenance	Experimental role
GRASCO	Gate-aware Renewable-Alignment Stochastic Co-Optimization	proposed in this paper	complete method
GCJS	gate-conditioned joint Gaussian-copula scenarios	proposed in this paper	market-forecast component
WLPF	weather-load VRE-above-load proxy forecast	proposed in this paper	renewable-signal component
PERS	previous-day persistence [1]	standard naive forecast	point baseline
CLIM	rolling climatology [1]	standard naive forecast, present window	probabilistic baseline
LEAR	Lasso-estimated autoregression [2]	published forecast method	point baseline
LGBM	Light Gradient Boosting Machine [3]	published learning method	point baseline
EA-PERS / PT-RES	persistence energy arbitrage / threshold reserve [1]	present heuristic constructions	operational baselines
QR-Ind	independent quantile-regression scenarios [4]	standard method, present implementation	dependence ablation
Hist-P / Hist-I	paired / target-independent historical scenarios [11]	present baseline construction	simple scenario baselines
EC / PGC	empirical / pooled Gaussian copula [5]	standard copulas, present implementation	dependence comparators
PGC-NoCross / PGC-NoTime	PGC without cross-target / temporal dependence [5]	present ablations	component ablations
Gate-LGBM / WX-PGC	noon LGBM update / weather-augmented PGC [3, 5]	present information sensitivities	added-information tests
DD-CC	data-driven chance-constrained scheduling [13]	DK1 reconstruction	external comparator
MPF	multi-phase stochastic framework [14]	DK1 reconstruction	external comparator
W&SPP	wind and storage power-plant bidding [15]	DK1 reconstruction	external comparator
PI	perfect-information dispatch [12]	infeasible-information reference	upper bound

Table 6: Pre-specified experimental protocol.

Item	Frozen setting
Market evaluation	211 chronological delivery days beginning 28 January 2026
Weather-common evaluation	209 days with complete published wind/solar inputs
WLPF evaluation	208 days and 4,990 valid hours per gate
GRASCO matched evaluation	206 days complete at both 07:30 and 12:00 gates
Forecast windows	84-day fit, 28-day probability calibration, 7-day clearing gap
Battery	4 MW / 8 MWh; round-trip efficiency 0.85; initial and terminal SOC 4.4 MWh
Scenarios	100 generated, 50 equal-weight scenarios supplied to optimization
Randomness	seeds 1, 2, and 3 for forecast scores; seed 1 for decision export
Risk	lower-tail CVaR level 0.95; $\chi \in \{0, 0.5\}$, with W&SPP also at 0.1
GRASCO sensitivity	$\rho \in \{0, 25, 50, 100\}$ EUR/MWh; realized profit excludes the credit
Missingness	preserve missing values; no zero fill, interpolation, or extrapolation
Uncertainty	six-lag DM test and 2,000-repetition seven-day moving-block bootstrap

and decision, risk, proxy, and linkage outputs are stored in `s3_comparison`. Each directory’s summary file records the environment, random seeds, configuration hash, and output hashes.

Each reported result has the lineage in Table 7; the table separates descriptive proxy evidence from optimizer and settlement outputs.

Table 7: Formula and data lineage of the empirical results.

Result block	Immediate inputs	Producing equations
Proxy occurrence and energy	observed wind, solar, load	Eqs. (13)–(16)
Export and Power-to-Heat explanation	proxy, exchange, non-VRE generation, Power-to-Heat	Eqs. (20)–(26)
Gate forecast scores	gate weather, published lags, realized proxy	Eqs. (41)–(51)
GRASCO decision sensitivity	WLPF, PGC market scenarios, planned charging	Eqs. (69)–(72)
Strategy settlement outcomes	market scenarios, physical constraints, realized prices and activation	Eqs. (54)–(80)
Midday–evening alignment	deployed charge, realized proxy, realized SOC and reserve	Eqs. (27)–(87)
Shortfall attribution	required and delivered activation, power and SOC traces	Eqs. (62)–(84)

13 Results

13.1 Market point forecasts

Table 8 reports the main 211-day point-forecast comparison. LGBM has the lowest MAE for the two price targets for which all three learned reference models are available. The frozen LGBM supplies all four decision targets and improves activation-volume and activation-price MAE over PERS, but it is not uniformly best for day-ahead or capacity prices.

Table 8: Point-forecast MAE on 211 delivery days (authors’ calculations). LGBM follows Ke et al. [3]; LEAR follows Ziel and Weron [2].

Method	Day-ahead price EUR/MWh	Capacity price EUR/MW	Activation volume MWh/h	Activation price EUR/MWh
PERS	28.875	9.080	13.573	109.312
Frozen LGBM	30.104	9.832	10.260	83.371
LEAR	25.458	12.272	–	–
LGBM	24.842	8.812	–	–

13.2 Marginal and joint probabilistic forecasts

CRPS and Energy Score are negatively oriented. Table 9 shows that no method dominates every marginal: GCJS is best for day-ahead and capacity price, the rolling climatology is best for activation volume, and EC is best for activation price. The purpose of GCJS is the joint path rather than four independently optimized marginal rankings.

Table 10 scores the same methods jointly across all four market targets rather than one target at a time. MPF and GCJS are statistically tied at the top, both clearly ahead of the empirical

Table 9: Mean sample CRPS by target on the 211-day evaluation set (authors’ calculations).

Method	Day-ahead price	Capacity price	Activation volume	Activation price
Rolling climatology	27.231	9.222	7.712	64.290
QR-Ind	21.556	7.463	7.839	63.070
Hist-P	27.179	9.213	7.730	64.251
Hist-I	27.270	9.259	7.719	64.299
EC	21.674	7.480	7.879	62.488
GCJS	21.499	7.415	7.842	63.019
PGC-NoCross	21.654	7.448	7.841	63.175
PGC-NoTime	21.555	7.455	7.841	63.014
MPF scenarios [14]	21.616	7.438	7.877	62.528

Table 10: Joint Energy Score for the full market vector on 211 days (authors’ calculations; lower is better).

Method	Energy Score	Method	Energy Score
MPF scenarios [14]	5.26173	GCJS	5.26174
PGC-NoCross	5.27024	PGC-NoTime	5.27291
QR-Ind	5.29161	EC	5.29868
Hist-I	5.89988	Rolling climatology	5.91644
Hist-P	5.93666		

and climatology comparators, and removing cross-target or temporal dependence (PGC-NoCross, PGC-NoTime) worsens the score relative to GCJS. The paired tests in Table 11 prevent the tiny numerical difference between MPF and GCJS from being over-interpreted. GCJS significantly improves on independent quantile scenarios, but does not significantly beat EC or MPF. Removing cross-target or temporal dependence worsens the point estimate of the joint score, so those structures contribute in the expected direction, but the decision ablations below test whether the forecast improvement has economic value.

Table 11: Pre-specified paired forecast comparisons (method minus comparator; authors’ calculations).

Method	Comparator	Mean loss difference	p -value
GCJS	QR-Ind	−0.03588	0.00162
GCJS	EC	−0.03349	0.05843
GCJS	MPF scenarios [14]	−0.00632	0.72668
WX-PGC	GCJS, common 209 days	+0.20724	0.03972
12:00 conditional GCJS	Unconditional GCJS	+5.88920	0.000002

The noon update is a negative result. Unconditional GCJS has CRPS 21.499, whereas conditioning its Gaussian copula on the published capacity path gives 27.351. Likewise, the day-ahead-price MAE of the LGBM reference rises from 24.842 to 31.841 after the implemented gate update. The WX-PGC information sensitivity also worsens the common-sample joint score from 5.269 to 5.468. Later or additional information therefore does not compensate for a poorly calibrated update map in this sample.

13.3 Decision value and external-method comparison

Table 12 reports the risk-neutral results on the 211-day common market sample. Profit and realized delivery are evaluated by the same settlement ledger for every computed arm. PI uses unavailable future information and is an upper bound, not a competitor. DD-CC, MPF, and

W&SPP are the DK1 reconstructions defined above.

Table 12: Main decision results under the common DK1 protocol (authors’ calculations).

Method	Mean profit EUR/day	5% profit CVaR EUR/day	Undelivered activation %	Recovery cost EUR/day
EA-PERS	825.0	−39.3	0.00	–
PT-RES	2,055.1	183.6	7.38	–
LGBM-DO	2,687.6	643.3	12.86	485.1
Hist-P-SO	3,018.2	759.4	8.83	231.4
GCJS-SO	3,117.8	805.5	10.62	255.7
PGC-SO	3,133.8	793.5	10.52	258.3
DD-CC reconstruction [13]	2,963.5	751.2	11.90	324.0
MPF reconstruction [14]	3,113.9	766.7	9.66	245.1
W&SPP reconstruction [15]	2,917.9	641.0	11.80	223.2
PI upper bound	3,908.7	1,064.9	0.00	–

Hist-P-SO improves mean realized profit over LGBM-DO by EUR 330.6/day, with a 95% block-bootstrap interval of [199.6, 479.2] EUR/day, and reduces the daily undelivered-activation share by 1.462 percentage points, with interval [−2.627, −0.450]. GCJS-SO improves profit over the same point baseline by EUR 430.3/day ([277.2, 588.8]), but its change in undelivered share is not significant. PGC-SO produces the highest mean profit among the risk-neutral internally built market methods, but its EUR 31.6/day advantage over pooled independent scenarios ($p = 0.262$) and EUR 48.5/day advantage over PGC-NoCross ($p = 0.161$) are not statistically resolved. The supported decision claim is therefore the value of probabilistic scenarios over point dispatch, not the universal superiority of one copula.

13.4 Risk-weight sensitivity

Table 13 repeats the profit, tail-risk, and delivery columns of Table 12 for the scenario methods at a positive fixed CVaR weight χ , to test whether raising χ trades mean profit for a calibrated improvement in realized lower-tail outcomes.

Table 13: Fixed CVaR-weight sensitivity (authors’ calculations).

Method	χ	Mean profit EUR/day	5% profit CVaR EUR/day	Undelivered activation %
Hist-P-SO	0.0	3,018.2	759.4	8.826
Hist-P-SO	0.5	2,992.9	766.7	8.842
GCJS-SO	0.0	3,117.8	805.5	10.624
GCJS-SO	0.5	3,080.9	788.6	10.612
PGC-SO	0.0	3,133.8	793.5	10.517
PGC-SO	0.5	3,085.7	778.8	10.607
W&SPP [15]	0.0	2,917.9	641.0	11.798
W&SPP [15]	0.1	2,868.9	664.0	11.325
W&SPP [15]	0.5	2,480.4	595.4	10.105

The fixed risk weight does not consistently improve either realized lower-tail profit or delivery. Across scenario methods, the realized exceedance rate of the nominal 5% profit threshold ranges from approximately 6.2% to 31.9%, and reaches 44.4% among all tested configurations. Model-internal CVaR is therefore not a calibrated guarantee of ex-post delivery loss.

13.5 Data coverage and proxy occurrence

The proxy data cover 1 October 2025 through 27 August 2026. There are 330 complete delivery days and 7,920 valid hourly observations. The missing day 25 August 2026 and the incomplete 28 August 2026 observation are excluded from the proxy statistics. Forecast scores use 208 delivery days and 4,990 hours per gate. Because 5 and 12 April each lack one lagged predictor hour, the GRASCO comparison uses the 206 days with complete 24-hour inputs at both gates; no missing hour is filled.

Table 14: Observed occurrence of the DK1 VRE-above-load proxy (authors’ calculations from Energinet settlement data [16]).

Sample	Positive hours	Valid hours	Positive-hour share
All complete hours	2,097	7,920	26.48%
Midday, 10:00–15:59	565	1,980	28.54%
July 2026	324	744	43.55%
August 2026 complete days	244	624	39.10%
Local hour 16:00	103	330	31.21%
Local hour 17:00	103	330	31.21%

The full-sample proxy energy is 1,580,970 MWh, of which 427,646 MWh occurs in the fixed midday window. Table 14 shows that the phenomenon is seasonal but not confined to solar noon. DK1 wind generation creates a broad surplus-proxy pattern, with the highest clock-hour frequency at 16:00–17:00. The empirical object is therefore a renewable-rich midday-to-evening transition, not a purely solar “duck-curve” interval.

13.6 What explains the 2,097 proxy-positive hours?

All 2,097 proxy-positive hours are net-export hours and also contain positive non-VRE generation and Power-to-Heat consumption. Table 15 reports quantities only on those hours, so its columns are accounting components rather than mutually additive estimates of curtailment.

Table 15: Accounting decomposition on the 2,097 proxy-positive hours (authors’ calculations from Energinet settlement data [16]).

Quantity	Aggregate value
VRE-above-load proxy, X^{proxy}	1,580,970 MWh
Net export, E	2,811,562 MWh
Non-VRE generation, C	1,230,592 MWh
Power-to-Heat already included in load, H	679,051 MWh
Export coverage, EC	100.00%
Non-VRE share of net export, CS	43.77%
Power-to-Heat counterfactual share, HS	30.05%
Maximum absolute hourly balance residual	1.6×10^{-5} MWh

The measured balance gives $2,811,562 - 1,230,592 = 1,580,970$ MWh up to numerical precision. Consequently, the original proxy is the portion of net exports left after accounting for non-VRE generation; it is not energy shown to be unused in DK1. If Power-to-Heat were removed only as an accounting counterfactual, the proxy on these hours would rise to 2,260,021 MWh. Observed Power-to-Heat therefore absorbs 679,051 MWh, or 30.05% of that counterfactual quantity, and must not be deducted a second time from gross consumption.

13.7 Gate-available proxy forecast results

The forecast evaluation contains 4,990 hours at each gate across 208 delivery days. The spring daylight-saving day, the incomplete final settlement day, and 24 weather-feature omissions on 24 June are not evaluated; the 7- and 14-day lag construction additionally leaves one local hour missing on 5 and 12 April. No missing value is filled. The minimum weather-availability margin before either decision gate is four hours. Table 16 reports the WLPF component errors and the resulting proxy classification skill (Eqs. (47)–(50)) at both gates, over all hours and over the midday window alone.

Table 16: Gate-available WLPF component and proxy forecast performance (authors’ calculations from Energinet and ECMWF/Open-Meteo data [16, 23]).

Gate	Scope	Wind MAE	Solar MAE	Load MAE MWh	Proxy MAE	Proxy RMSE	Precision %	Recall
07:30	All hours	408.8	238.8	224.7	180.4	410.3	79.19	61.92
12:00	All hours	377.7	231.4	221.5	174.9	403.8	80.66	66.10
07:30	Midday	468.3	506.6	278.1	276.4	562.0	61.92	54.20
12:00	Midday	445.3	490.0	270.9	270.3	553.2	64.11	59.54

Moving from 07:30 to 12:00 reduces all-hour proxy MAE by 3.06% and RMSE by 1.59%; midday reductions are 2.20% and 1.57%, respectively. The mean absolute gate-to-gate proxy revision is 41.98 MWh, and the positive-hour classification changes in 243 of 4,990 common hours (4.87%). The later run therefore adds information, although the update is incremental rather than a regime change.

The forecasts are conservative in energy magnitude. Against 1,143,143 MWh of observed proxy energy, the 07:30 and 12:00 models predict 669,964 MWh (58.61%) and 724,365 MWh (63.37%), respectively. In the midday window the corresponding shares are 71.03% and 76.23%. The signal is sufficiently precise to define a soft GRASCO reward or diagnostic, but its recall and energy underprediction do not support treating it as a hard estimate of all renewable-rich hours.

13.8 Midday–evening outcomes by method

Table 17: Market-only mean daily linkage outcomes on 206 common forecast-complete days (authors’ calculations).

Strategy	Midday charge (MWh/day)	SOC at 17:00 (MWh)	Evening reserve (MW h/day)	Evening delivery rate (%)	Recovery cost (EUR/day)
LGBM-DO	6.212	6.881	18.463	97.38	481.5
Gate-LGBM sensitivity	6.108	6.850	18.463	97.38	471.6
Hist-P-SO, $\chi = 0$	1.035	6.520	19.190	97.85	221.8
Hist-P-SO, $\chi = 0.5$	1.118	6.629	18.706	97.81	217.3
PGC-SO, $\chi = 0$	1.920	6.722	18.064	98.23	255.7
PGC-SO, $\chi = 0.5$	2.404	6.872	17.685	98.20	258.5
DD-CC reconstruction [13]	2.695	6.884	18.648	97.68	321.6
MPF reconstruction [14]	2.003	6.767	18.271	98.56	234.3
W&SPP reconstruction [15]	0.513	5.971	19.612	97.51	215.9

The five-hour evening window has a physical maximum of $4 \text{ MW} \times 5 \text{ h} = 20 \text{ MW h/day}$. The observed commitments in Table 17 therefore occupy 88.4–98.1% of this power-duration envelope. Requested evening activation is only 8.1–8.3% of committed reserve energy. Additional midday energy cannot materially increase reserve capacity when the power commitment is already near 4 MW; its primary physical value is to raise the pre-peak SOC and protect delivery.

The LGBM-DO schedules perform the most midday charging, but their recovery purchases cost approximately EUR 472–482/day. The scenario-based methods reduce this to EUR 216–322/day. The point-forecast schedules recover less energy than several stochastic methods but pay approximately EUR 54/MWh of recovered energy, compared with EUR 22–32/MWh for the scenario-based strategies. Its weakness is therefore recovery timing, not merely recovery volume.

The W&SPP reconstruction [15] is the clearest reserve-oriented method: it commits 19.62 MW h/day in the evening but charges only 0.51 MWh/day at midday. The MPF reconstruction [14] has the highest evening delivery rate, 98.56%, while retaining moderate midday charging and recovery cost. PGC-SO at $\chi = 0$ gives the highest mean daily total settlement value among the displayed methods, but its higher midday charging relative to historical pairing is accompanied by lower evening reserve and delivered activation. Increasing χ from zero to 0.5 raises midday charging and 17:00 SOC in the PGC-SO case, but does not improve the evening delivery rate or reserve commitment.

13.9 GRASCO renewable-alignment sensitivity

Table 18 compares GRASCO-0, the PGC-SO market-only control and $\rho = 0$ member of Eq. (72), with the three positive alignment credits. All four rows use the same 206 delivery days, market scenarios, battery constraints, realized settlement, and $\chi = 0$; their configured optimization problems differ only in ρ . Under the pre-specified persistence fallback for an unsuccessful numerical solve, the $\rho = 0, 25, 50, 100$ arms fall back on 3, 1, 2, and 0 days, respectively; those days remain in the matched-sample results. The fallback is a solver safeguard and receives no artificial alignment payment in the realized settlement ledger. Forecast overlap is evaluated against the 12:00 signal used by the deployed energy plan, whereas realized overlap uses X^{proxy} after delivery.

Table 18: GRASCO sensitivity on 206 common forecast-complete days (authors’ calculations; GRASCO is proposed in this paper).

Method	Midday charge	Forecast overlap MWh/day, except reserve in	Realized overlap MW h/day	SOC at 17:00	Evening reserve	Delivery (%)	Realized profit change (EUR/day)
GRASCO-0	1.920	0.615	0.775	6.722	18.064	98.23	0.0
GRASCO-25	2.441	1.141	1.194	6.784	18.014	98.24	+13.4
GRASCO-50	3.418	2.112	1.815	6.869	17.948	98.24	−18.0
GRASCO-100	5.232	4.038	2.951	6.967	17.955	98.24	−71.5

Relative to GRASCO-0, GRASCO-25 adds 0.521 MWh per day of midday charge, 0.526 MWh per day of forecast overlap, 0.419 MWh per day of realized overlap, and 0.062 MWh of 17:00 SOC. Its paired mean profit difference is EUR +13.4 per day, with a 95% interval of [2.7, 28.2] EUR per day. Under GRASCO-50, realized overlap rises by 1.040 MWh per day but the profit difference is EUR −18.0 per day with interval [−42.9, 10.0]. Under GRASCO-100, realized overlap rises by 2.176 MWh per day and the profit difference is EUR −71.5 per day with interval [−115.6, −24.3]. These are sensitivity results, not evidence that EUR 25/MWh is a generally optimal social value.

The corresponding changes in evening reserve are −0.050, −0.116, and −0.109 MW h/day, and none is statistically distinguished from zero at 5%. Changes in delivered evening activation are also not significant. The implemented forecast link therefore changes charge timing and stored energy, but the already-frozen and power-limited evening reserve quantity remains largely unchanged.

13.10 Diagnosis of the observed problem

On the 206 matched decision days, the midday proxy contains 318,957 MWh. Among the nine market-only strategies, the largest overlap is 449.5 MWh for the LGBM-DO. Their maximum

observed proxy-absorption ratio is

$$\max_m AR_m^{\text{proxy}} = \frac{449.5}{318,957} = 0.001409 = 0.1409\%. \quad (88)$$

Equation (88) is the full-sample aggregation of the daily numerator in Eq. (28) divided by the matched-day proxy-energy denominator in Eq. (30); it is not an average of daily percentages. For the scenario-based and published-method market-only reconstructions, the corresponding range is 0.0116–0.0698%. Their realized alignment share is 34–44%. GRASCO-25 through GRASCO-100 raise that share to 48.9–56.4%, and GRASCO-100 reaches

$$AR_{\rho=100}^{\text{proxy}} = \frac{608.0}{318,957} = 0.001906 = 0.1906\%. \quad (89)$$

Comparing Eq. (89) with Eq. (88) shows that the implemented signal changes timing in the intended direction, but even the largest sensitivity remains below 0.2% of midday proxy energy.

The observed linkage statistics are

$$\Gamma_S \in [0.33, 0.52], \quad \Gamma_R \in [-0.52, -0.28], \quad DR^{\text{eve}} \in [97.38\%, 98.56\%]. \quad (90)$$

The three ranges in Eq. (90) come directly from Eqs. (86), (87), and (84), respectively. The correlations are descriptive rather than causal. Their joint pattern is nevertheless informative: midday charging consistently raises the 17:00 SOC, but days with more midday charging do not receive more evening reserve. This is consistent with the frozen gate sequence and the near-saturation of reserve power, rather than with a failure of the battery to carry energy from midday to the evening.

Across the full day, 74–88% of undelivered activation is attributable to gross-power headroom, while only 12–26% is attributable to the SOC-energy envelope. Evening shortfall averages only 0.021–0.040 MWh/day. Thus the remaining delivery problem is mainly simultaneous use of battery power by the energy schedule and activation, not inadequate evening SOC. Table 19 summarizes the evidence for and against each of the five candidate explanations for a shallow midday–evening link.

Table 19: Evidence-based diagnosis of the five candidate explanations.

Candidate explanation	Evidence-based diagnosis
Insufficient midday charging	Sensitive to the proxy credit. Market-only advanced methods charge 0.51–2.70 MWh/day. Proxy-aware charging reaches 5.23 MWh/day, but proxy absorption remains at most 0.1906%.
Excessive SOC reserved for aFRR	Ex-post slack, not the main bottleneck. SOC at 17:00 is 5.97–6.97 MWh, but the one-hour reserve requirement is 4.339 MWh and the energy constraint rarely binds.
Inadequate evening delivery	Secondary. Aggregate delivery is 97.38–98.56%; average evening shortfall is at most 0.040 MWh/day.
Excessive activation-recovery cost	Material for point-forecast dispatch; reduced by stochastic scheduling. The cost is EUR 474–483/day for point-forecast dispatch versus EUR 215–323/day for scenario methods.
Midday charging and evening reserve are already coordinated	Forecast link changes energy, not reserve. Positive ρ raises charge and SOC, but evening reserve changes are not significant because it is fixed at the earlier gate and near its power ceiling.

14 Research claim and remaining uncertainty

The empirical claim of the present DK1 study is:

For a price-taking 4 MW/8 MWh battery participating in day-ahead energy and upward aFRR, probabilistic scheduling improves reserve value, activation delivery, and recovery timing, but the market-only schedules produce little charging aligned with the DK1 VRE-above-load proxy. Supplying the gate-safe WLPF signal to the proposed GRASCO alignment objective increases forecast and realized overlap, mid-day charging, and 17:00 SOC monotonically with ρ . It does not materially change evening reserve or delivered activation because reserve is fixed at 07:30 and already close to the battery power limit. The profit response is non-monotone: GRASCO-25 improves mean realized profit on this sample, GRASCO-50 is statistically inconclusive, and GRASCO-100 lowers profit. These proxy-positive observations represent renewable-attributable net-export pressure rather than measured curtailment.

This claim is narrower than a claim of actual curtailment reduction. It is useful to state the decision chain implemented only by the $\rho > 0$ arms:

$$\hat{X}_{d,h}^{\text{proxy},07:30} \rightarrow r_{d,h}^{\text{up}}, P_{d,c,t}^{\text{ch}} \rightarrow \hat{X}_{d,h}^{\text{proxy},12:00} \rightarrow P_t^{\text{ch}} \rightarrow S_{17} \rightarrow e_{17:21}^{\text{del}} \rightarrow C^{\text{rec}}, \quad (91)$$

GRASCO-0 sets $\rho = 0$, so WLPF has no objective value; the three positive sensitivity arms implement Eq. (91) through Eqs. (69)–(72). The realized proxy X^{proxy} still enters only ex-post diagnostics. The remaining uncertainty is not whether the signal can change charge timing, which the experiment establishes, but how the economic trade-off varies with forecast quality, valuation, battery scale, and unobserved system curtailment.

The numerical evidence is stored in the proxy-surplus, decomposition, and gate-forecast tables, the daily and summary midday–evening tables, and the nine market-only and three GRASCO renewable-alignment dispatch traces. The experiment result manifest records the exact output names. The trace-to-daily aggregation error is at most 5.14×10^{-8} , confirming that the new ledger reproduces rather than changes the frozen settlement results.

15 Aggregation rules and admissible claims

Metrics are evaluated only on common intervals for which all required variables are observed. Missing values are not replaced by zeros, interpolated, or extrapolated. For multi-day or full-sample results, ratios are computed as the sum of their numerators divided by the sum of their denominators; daily ratios are not averaged without weights. If the denominator of Eq. (29) or Eq. (30) is zero, the corresponding daily metric is recorded as missing rather than zero.

The primary terminology for the present study is “charging aligned with renewable-rich midday periods,” “potential renewable absorption,” and “midday-to-evening energy shifting.” The terms “actual renewable utilization rate” and “actual curtailment reduction” require unconstrained available VRE, observed curtailment or dispatch-down information, and a system-level counterfactual market-clearing model.

16 Relation to established formulations

Equations (7)–(10) follow the common system-dispatch distinction between available VRE and utilized VRE. The National Renewable Energy Laboratory (NREL) represents curtailment as the unused portion of otherwise available variable generation and evaluates storage within the system balance [20]. Its formulation for DC-coupled PV–battery systems further tracks curtailment recovered by storage charging within an explicit energy balance [21]. Peng et al. jointly model generation, curtailment, transmission, and storage operation in SWITCH-China and constrain renewable-connected batteries to charge from co-located renewable generation [22]. These studies can evaluate system curtailment or renewable charging provenance because their models contain system balances, available generation, or source-coupling constraints that are absent from the present price-taking DK1 simulation.

Table 20: Permitted and prohibited interpretations.

Metric or evidence	Permitted interpretation	Prohibited interpretation
Eq. (9) with complete inputs Eq. (11)	Curtailment rate under the stated system boundary Model-based reduction under a controlled system counterfactual	Replacing available generation with realized generation A price-taking DK1 battery changed observed market clearing
Eq. (27)	Simulated charging in the fixed midday window	Actual renewable energy utilized at midday
Eqs. (28)–(30)	Temporal alignment with the local VRE-above-load proxy	Every charged MWh came from VRE or reduced curtailment one-for-one
Eqs. (21)–(26)	Accounting decomposition of proxy-positive hours	Exports or Power-to-Heat are equivalent to observed curtailment
Eq. (46)	Model forecast of the reproducible ex-post proxy	Official TSO load forecast or physical-curtailment forecast
Eq. (32)	Simulated charging at a high VRE-to-load ratio	Curtailment necessarily occurred above the threshold
Eq. (33)	Simulated charging during negative-price intervals	Negative prices are equivalent to renewable curtailment

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